

Clinical Variant Interpretation Report

Report metadata

Report schema: 1.0 Evidence schema: 1.0 Interpretation prompt: 1.1 LLM model: gemini-3.1-flash-lite Assembly: GRCh38

Case Summary

- Provided HPO terms: HP:0002315
- Matched HPO terms: HP:0002315

Variant Summary

- Assembly: GRCh38
- Variant: 3:37048991 G>A
- Population frequency: Not available in the supplied evidence.

Gene and Consequence

- Gene: MLH1
- Gene ID: ENSG00000076242
- Transcript: ENST00000231790
- Consequence: missense_variant
- Impact: MODERATE
- Protein change: ENSP00000231790.3:p.Glu693Lys

Clinical Evidence

ClinVar

- Accession: Not available in the supplied evidence.
- Clinical significance: Not available in the supplied evidence.
- Review status: Not available in the supplied evidence.
- Conditions: Not available in the supplied evidence.

ClinGen

- Disease: Lynch syndrome; Disease ID: MONDO:0005835; Classification: Definitive; Mode of inheritance: Autosomal dominant
- Disease: mismatch repair cancer syndrome 1; Disease ID: MONDO:0010159; Classification: Definitive; Mode of inheritance: Autosomal recessive
- Disease: hereditary breast carcinoma; Disease ID: MONDO:0016419; Classification: Refuted Evidence; Mode of inheritance: Autosomal dominant

Phenotype Correlation

- Phenotype score: 1
- Provided HPO terms: HP:0002315
- Matched HPO terms: HP:0002315

Interpretation

The provided evidence identifies the *MLH1* gene as having a definitive role in Lynch syndrome and mismatch repair cancer syndrome 1. However, the evidence does not contain a specific functional or clinical classification for the *MLH1* p.Glu693Lys variant itself.

Limitations

- No specific variant-level classification (e.g., pathogenic/benign) is available in the provided evidence.
- NCBI ClinVar did not return an exact result for this variant.
- Population frequency data is not available.

Evidence source availability

- vep: success
- myvariant: success
- clinvar: not_found
- clingen: success

Report warnings

- NCBI ClinVar returned no exact result for this variant.

References

- Ensembl VEP - https://rest.ensembl.org/vep/homo_sapiens/region
- MyVariant.info - <https://myvariant.info/v1/variant/chr3%3Ag.37048991G%3EA?assembly=hg38>
- ClinGen Gene-Disease Validity via UCSC GenCC - https://search.clinicalgenome.org/kb/gene-validity/CGGV:assertion_24c5d3f4-2bb6-4beb-a781-52bd6962ca41-2022-06-20T17:00:00.000Z
- ClinGen Gene-Disease Validity via UCSC GenCC - https://search.clinicalgenome.org/kb/gene-validity/CGGV:assertion_a313ad2e-5c2a-4bb4-bed4-3b4f9b5095df-2018-10-09T15:06:23.105Z
- ClinGen Gene-Disease Validity via UCSC GenCC - https://search.clinicalgenome.org/kb/gene-validity/CGGV:assertion_c27a22e4-fc45-4888-8b1d-625bc649697c-2023-12-21T18:00:00.000Z
- ClinGen - MONDO:0005835 - https://search.clinicalgenome.org/kb/gene-validity/CGGV:assertion_24c5d3f4-2bb6-4beb-a781-52bd6962ca41-2022-06-20T17:00:00.000Z
- PubMed - PMID:8673133
- PubMed - PMID:10096563
- PubMed - PMID:18566915
- PubMed - PMID:24802709
- PubMed - PMID:33335961
- ClinGen - MONDO:0010159 - https://search.clinicalgenome.org/kb/gene-validity/CGGV:assertion_a313ad2e-5c2a-4bb4-bed4-3b4f9b5095df-2018-10-09T15:06:23.105Z
- PubMed - PMID:9500552
- PubMed - PMID:9927033
- PubMed - PMID:9927034
- PubMed - PMID:10763829
- PubMed - PMID:11389087

- PubMed - PMID:11809679
- PubMed - PMID:14762794
- PubMed - PMID:15139004
- PubMed - PMID:15340263
- PubMed - PMID:17440981
- PubMed - PMID:18593904
- PubMed - PMID:22692065
- ClinGen - MONDO:0016419 - https://search.clinicalgenome.org/kb/gene-validity/CGGV:assertion_c27a22e4-fc45-4888-8b1d-625bc649697c-2023-12-21T18:00:00.000Z
- PubMed - PMID:28418444
- PubMed - PMID:33471991
- PubMed - PMID:35172496

Medical Disclaimer

AI-generated decision-support summary based only on the supplied Evidence Object. It is not a diagnosis or treatment recommendation and requires review by a qualified healthcare professional.